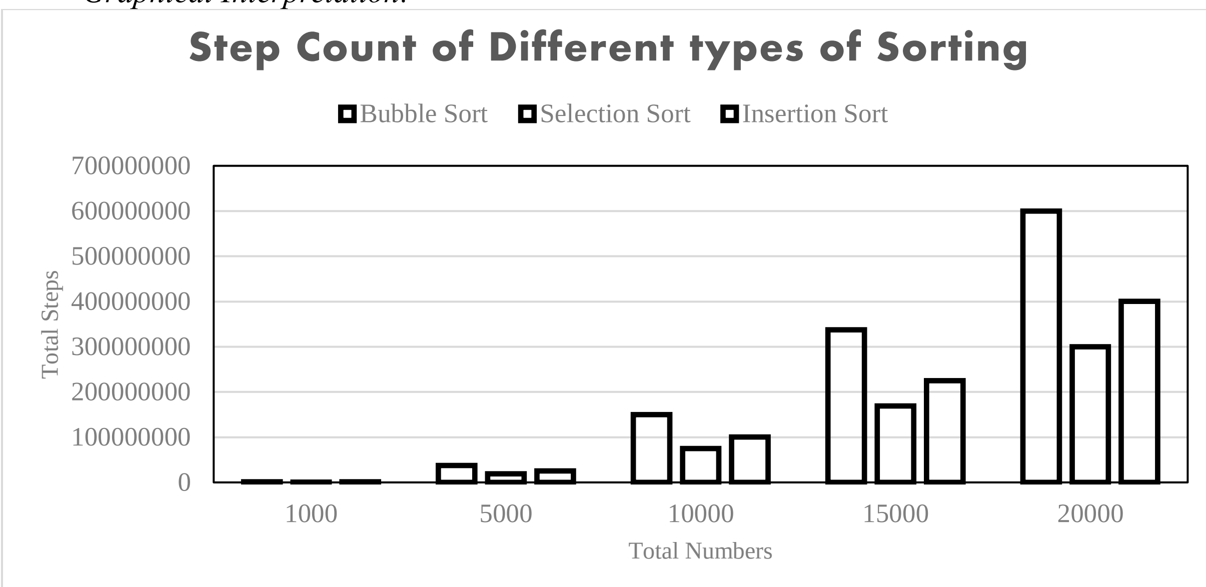
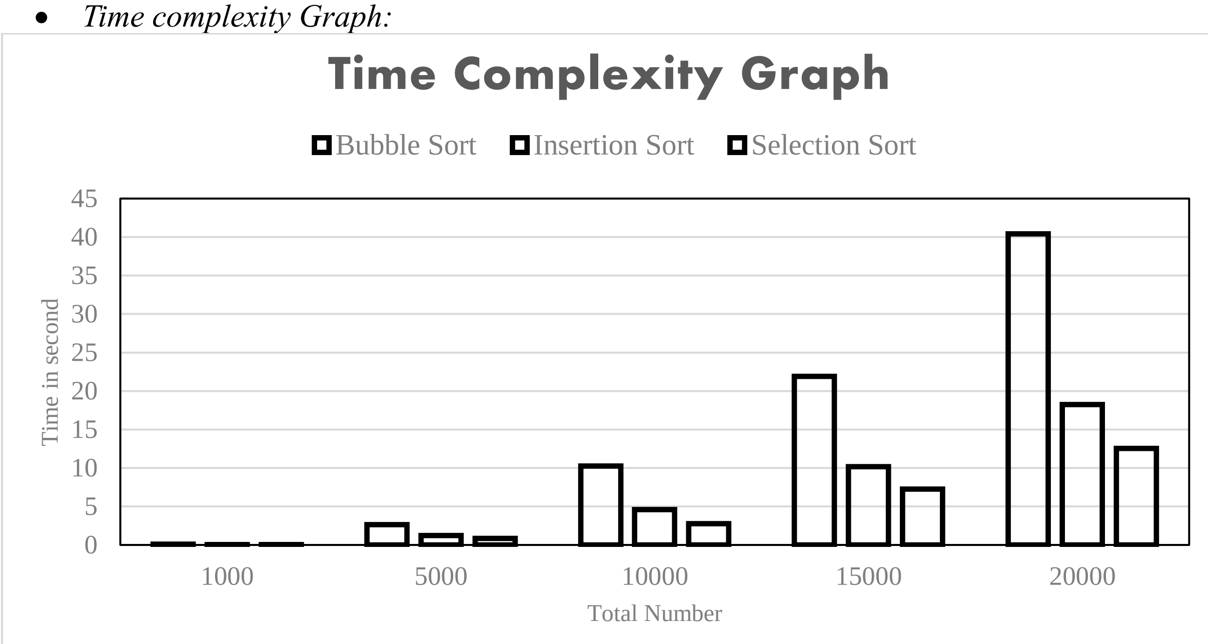
Output: (Steps)

No. of Data	Bubble Sort	Selection Sort	Insertion Sort
1000	1503502	754502	1003997
5000	37517502	18772502	25019997
10000	150035002	75045002	100039997
15000	337552502	168817502	225059997
20000	600070002	300090002	400079997

- *Graphical Interpretation:*



- *Time complexity Graph:*



- Implemented Code (C++):

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int

class Sorting_Algorithm
{
   public:
   ll Bubble_Sort(vector<int>&A)
   {
      ll step=0;
      int n=A.size();
      step++;
      for(int i=0;i<n;i++)
      {
         step++;
         bool flag=true;
         step++;
         for(int j=0;j<n;j++)
         {
            step++;
            if(A[j]>A[j+1])
            {
               step++;
               swap(A[j],A[j+1]);
               flag=false;
            }
         }
         step++;
         if(flag)
         {
            step++;
            break;
         }
      }
      step++;
      return step;

   }
   ll Insertion_sort(vector<int>&A)
   {
      int n=A.size();
      ll step=0;
      step++;
      for (int i=1;i<n;i++)
      {
         step++;
         int value=A[i];
         int j=i-1;
         step=step+2;
```

```
            while (j>=0 && A[j]>value)
            {
                step++;
                A[j+1]=A[j];
                j=j-1;
                step=step+1;
            }
            step++;
            A[j+1]=value;
            step++;
        }
        step++;
        return step;

    }
    ll Selection_Sort(vector<int>&A)
    {
        int n=A.size();
        ll step=0;
        step++;
        for(int i=0;i<n;i++)
        {
            step++;
            int mn=i;
            int j=i+1;
            step=step+2;
            while(j<n)
            {
                step++;
                if(A[j]<A[i])
                {
                    mn=j;
                    step++;
                }
                j++;
                step++;
            }
            step++;
            swap(A[i],A[mn]);
            step++;
        }
        step++;
        return step;
    }
};
void print(vector<int>A)
{
    for (int i:A)
    {
        cout<<i<<" ";
    }
```

```
    newl;
}

int main()
{
    vector<int>A[5];
    int B[5]={1000,5000,10000,15000,20000};
    for(int i=0;i<5;i++)
    {
        ifstream ip;
        ip.open("input.txt");
        for(int j=1;j<=B[i];j++)
        {
            int x;
            ip>>x;
            A[i].push_back(x);

        }
        ip.close();
    }
    Sorting_Algorithm sort=Sorting_Algorithm();
    vector<int>temp;
    ofstream op;
    op.open("output.txt");
    op<<"No. of Data\t Bubble Sort\t Selection Sort\t Insertion Sort\n";
    for(int i=0;i<5;i++)
    {
        op<<"  "<<B[i]<<"\t\t\t";
        temp=A[i];
        op<<sort.Bubble_Sort(temp)<<"\t\t";
        temp=A[i];
        op<<sort.Selection_Sort(temp)<<"\t\t";
        temp=A[i];
        op<<sort.Insertion_sort(temp)<<"\n";
    }
}
```